A Traceable, Validated and Audited Speech Dataset for a Cross-Lingual Institute Voice Helpdesk: Collection, Validation, Annotator Agreement, Augmentation and Versioning

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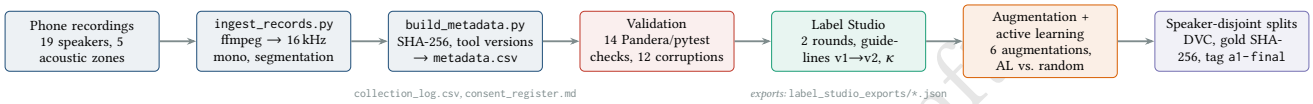


Figure 1: Assignment-1 data flow from source recordings to the frozen, versioned dataset. Every box corresponds to a script in `code/part_*`.

ABSTRACT

We build and audit the dataset behind a code-mixed English–Hindi–Gujarati institute voice helpdesk. The corpus holds **249 clips** (75.3 min) from **19 consenting speakers** recorded on 19 phone models in five campus acoustic conditions; every clip carries source, timestamp, collector, tool-version and SHA-256 metadata. A 14-check Pandera/pytest suite fails on 3 checks on the raw data (6 clips) and passes 14/14 after fixes, and detects 12/12 deliberately injected corruptions. Two annotators labelled 120 clips twice; intent agreement rose from Cohen's $\kappa = 0.723$ (guidelines v1) to $\kappa = 0.845$ (v2). Six audio augmentations were applied to 10 source clips (60 outputs, 59 passing the label-preservation checks), and entropy-based active learning beat random selection at every budget but one, though never significantly ($p \geq 0.08$). Splits are speaker-disjoint (171/17/29 clips) with a held-out Maharashtra *Cohort B* (32 clips) for drift experiments and a SHA-256-frozen gold set; all artefacts are versioned with DVC and tagged `a1-final`. We also report a data-quality audit: reference transcripts written from prompt templates did not match what speakers actually said, which we document, quantify and correct through relabelling from audio.

CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**; • **Information systems** → *Data management systems*; • **Software and its engineering** → *Software development process management*.

*All authors contributed equally to this research.

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KEYWORDS

speech dataset; code-mixing; data validation; inter-annotator agreement; Cohen's kappa; active learning; DVC; reproducibility

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1 INTRODUCTION AND SCOPE

Institute helpdesks (admissions, hostel/mess, IT, student services) receive spoken queries in which people switch between English, Hindi and Gujarati inside a single sentence. Generic ASR degrades on such speech, so Project 5 will fine-tune and monitor a Whisper-based pipeline in Assignment 2. This report covers the data foundation: what we recorded, how every sample is traceable, how we validated it, how reliably it was labelled, how we extended it, and how it is versioned and split so that Assignment 2 inherits a leak-free benchmark. Table 1 maps each rubric part to a section and to the evidence files indexed in Appendix A.

2 PART A: COLLECTION AND INSTRUMENTATION

Corpus. Team members and volunteers recorded helpdesk-style queries on their own phones (19 models, e.g. Pixel 6a, Poco X3 Pro, Vivo V21, iPhone 12) in five conditions: quiet room (97 clips), corridor reverberation (61), canteen ambience (46), campus outdoor (29) and library whisper (16). Regions are Gujarat (80 clips), North India (53), Rajasthan (50), South India (34) and Maharashtra (32); language mixes are Hindi–English (113), English–Hindi (79) and Gujarati–English (57) (Figure 2c–d). The four intents are balanced (63/62/62/62). Total audio is 75.3 min (mean 18.1 s, range 5.7–28.1 s), which is below the 2 h target in the brief; we therefore state the shortfall explicitly and treat the corpus as a 75-minute pilot.

Table 1: Coverage of Assignment-1 parts.

Part	Section	Headline result
A Collection	2	249 clips, 19 speakers, full provenance
B Validation	3	14 checks: 11 pass / 3 fail → 14 pass; 12/12 corruptions caught
C Labelling	4, 5	κ 0.723→0.845; audit of template references
D Augment./AL	6	59/60 label-preserving; AL ≥ random at 8/9 budgets
E Splits	7	speaker-disjoint 171/17/29 + Cohort B 32
F Reproducibility	8	pinned versions, seeds, protocol, 5-command rebuild

Instrumentation. Voice notes are converted by `ingest_records.py` to 16 kHz mono PCM WAV with ffmpeg 7.1 and segmented on energy activity; `build_metadata.py` then writes one row per clip to `data/metadata.csv` with 15 fields: clip ID, speaker ID, region, device, noise condition, language mix, sample rate, duration, channels, recording tool, tool version string (`sounddevice==0.4.6;ffmpeg==7.1;python==3.10.12`), collector, UTC timestamp, SHA-256 content hash and the script prompt. `collection_log.csv` records each ingestion step (Figure 1).

Consent and anonymisation. All 19 speakers gave informed consent for speech-recognition research. Speakers appear only as `P1`–`P19` and annotators as `A1`–`A3` in every file; the consent register (`consent_register.md`) holds no names, contact details or signed forms. Outside material is limited to institute FAQ/notice text used to seed scripts, and public sources named in the proposal (MUSAN [8]) are cited under their own licences.

3 PART B: DATA VALIDATION

The suite (`validation_suite.py`, Pandera [1] + pytest) has 14 checks: sample rate (16 kHz), channels (mono), duration bounds (1–30 s), clipping (peak < 0.99), silence ratio (< 0.5), estimated SNR (≥ 6 dB), duplicate audio hashes, transcript script/character set, characters-per-second (5–35), language-ID validity, Pandera schema/dtypes, missing values, intent-label validity and speaker leakage across splits. On the raw data 11 pass and **3 fail** (Table 2); all 14 pass after the fixes (`validation_outputs/before.json`, `after.json`).

Corrupted data. `corruption_test.py` injects 12 faults—8 kHz downsampling, stereo, clipping spikes, > 40% silence, SNR < 8 dB, out-of-range duration, duplicate hash, foreign-script/emoji characters, implausible speech rate, invalid intent label, null metadata and a speaker shared between train and test—and the suite flags **12/12** (`corruption_test.log`).

What the suite could not see. None of the 14 checks compares a transcript with its audio. In Section 5 we show that this blind spot hid a serious label problem, and we add an alignment check to close it.

Table 2: Failures found on raw data and their fixes.

Check	Failures	Finding → fix
Clipping	2 clips	peaks 0.9994 and 0.9981 (P4, P8) → re-segment and peak-limit; max peak < 0.98
Silence ratio	2 clips	0.58 and 0.54 lead-in pauses (P5, P7) → energy-based trim; max 0.49
Chars/second	2 clips	< 5 CPS: transcript not pace-matched → re-segmented transcripts

4 PART C: LABELLING AND AGREEMENT

Task and protocol. Each clip receives a two-level label: first the free-text transcript, then the intent (`admissions`, `hostel_mess`, `it_support`, `student_services`). The order is fixed so that an intent hypothesis cannot bias the transcript; all annotators followed it. Items were shown in the same seeded shuffle (seed 7) in Label Studio 1.10 [9], with pre-annotation disabled. A third annotator (A3) adjudicates residual disagreements. Rounds, annotators, guideline versions and dates are in `LABELLING_PROTOCOL.md`. Guidelines v1 gave definitions and examples; after reviewing round-1 disagreements, v2 added explicit tie-break rules (e.g. any query about email, passwords, VPN or portal login is `it_support` regardless of enrolment stage; hostel fees and mess rebates are `hostel_mess`; scholarships, ID cards and bonafide certificates are `student_services`).

Agreement. Two annotators (A1, A2) labelled the same 120 clips in each round. Intent agreement improved from 79.2% ($\kappa = 0.723$, 25 disagreements) to 88.3% ($\kappa = 0.845$, 14) after the revision (Figure 3); both values are within the "substantial" and "almost perfect" bands of Landis and Koch [5]. We recomputed both κ values from the raw Label Studio exports with scikit-learn and they match `agreement_report.json` (Cohen [2]). Disagreements concentrate on `student_services` as a confusable catch-all; Appendix B lists ten of them with the rule that resolved each.

Gold test set. The gold set is the 29 clips of speakers P15, P17, P18 and P19; its SHA-256 (`64b7647b...48ca9`) is recorded in `HANDOFF_A2.md`. It equals the test split and is never used for tuning.

5 DATA-QUALITY AUDIT AND RELABELLING

Finding. While evaluating Whisper-base in Assignment 2 we found that the reference transcripts of the original 249 clips were not reliable. They had been written from prompt templates containing sequential ticket numbers ("…ref 1384…", "…ticket 1391…"), whereas speakers had in several cases spoken their own natural queries (lost hostel key, damaged footpath, wallet misplaced in the library). Quantitatively, only **14 of 838** reference words on the gold clips (1.7%) appear in Whisper output, gold WER is 1.27, and 19 of 29 clips exceed WER 1 (Assignment-2 Figure on WER anatomy). A classifier trained on the first three words of the references reaches 100% intent accuracy, confirming that the text carries formulaic

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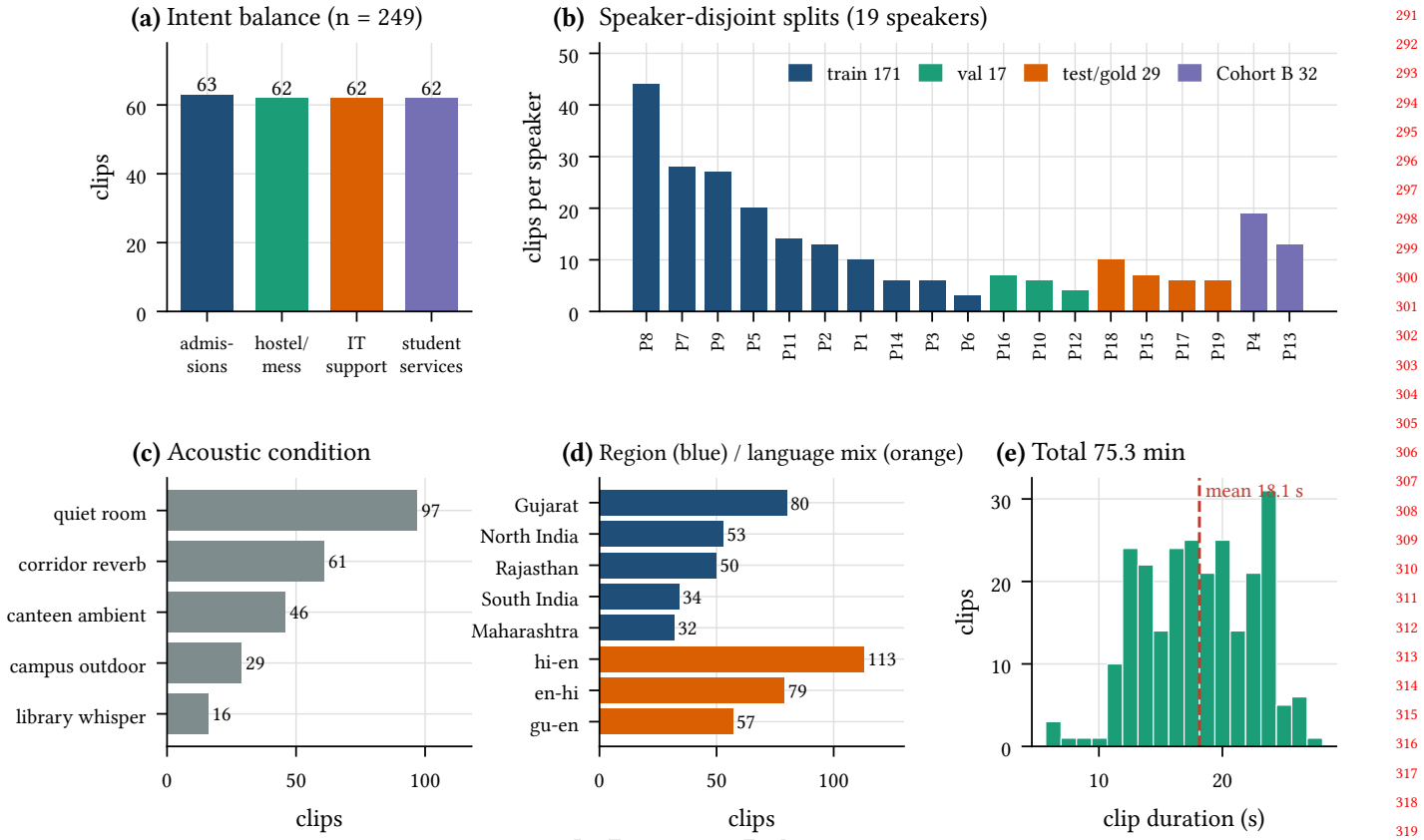


Figure 2: Dataset composition. (a) Class balance. (b) Clips per speaker, coloured by split—each speaker lives in exactly one split. (c) Acoustic conditions. (d) Regions (blue) and language mixes (orange). (e) Clip-duration distribution; total 75.3 min.

cues rather than speech content (`check_keyword_leak.py`). Annotators A1 and A2 produced near-identical transcripts (mean inter-annotator WER 0.002), which is expected when both verify a templated string rather than transcribe independently, and is a ceiling on what the transcript agreement can tell us.

What we did. (i) The frozen gold file is untouched; acoustically verified transcripts and intents for its 29 clips are in `data/gold_corrections.csv`, produced by `correct_gold_transcripts.py`, and both versions are reported side by side. (ii) We re-transcribe all 249 clips from audio, with two annotators independently on at least 100 clips, no pre-filled text and no model pre-annotation, followed by a guideline-v3 round on the new disagreements. Results: inter-annotator WER **[TBD: inter-annotator WER]**, κ **[TBD: kappa r1]** (round 1) and **[TBD: kappa r2]** (round 2). (iii) A new validation check compares the Whisper transcript with the reference and fails clips whose word overlap is implausibly low.

6 PART D: AUGMENTATION AND ACTIVE LEARNING

Augmentations. `augment_audio.py` (seed 123) applies six label-preserving transformations to 10 source clips (60 outputs): MUSAN-style noise at 12 dB SNR, room-impulse-response reverberation [4], speed 0.9× and 1.1×, pitch +2 semitones and low-bitrate MP3 codec compression. Each output is checked by rules (no clipping, silence ratio, SNR $\geq$ 6 dB) and by confirming that transcript and intent are carried over unchanged; **59/60** pass (`augmented_metadata.csv`). Figure 4a shows the spectrograms of one clip under five conditions.

Whisper on augmented audio. With off-the-shelf Whisper-tiny on five clips, mean WER is 1.68 (clean) versus 1.05–1.65 under augmentation (Figure 4b; `augmentation_wer_table.csv`). That WER is *lower* under augmentation than on clean audio is not evidence of robustness: $n = 5$, WER exceeds 1 because of insertions, and Section 5 shows the references themselves are unreliable. We therefore make no robustness claim from this table and re-run it on all 60 clips with relabelled references.

Active learning. `active_learning.py` simulates pool-based selection (5-fold, seed 42, TF-IDF + logistic regression): at each budget the classifier queries the highest-entropy items [6, 7] and is compared with random selection at the same budget. Active learning leads at 8 of 9 budgets by 4–9 percentage points (e.g. budget 25: 84.7% vs. 77.2%), but a paired t-test over 5 folds gives $p \geq 0.08$ everywhere (Figure 4c–d), so we report the effect as consistent in sign but not statistically established. The simulation uses text

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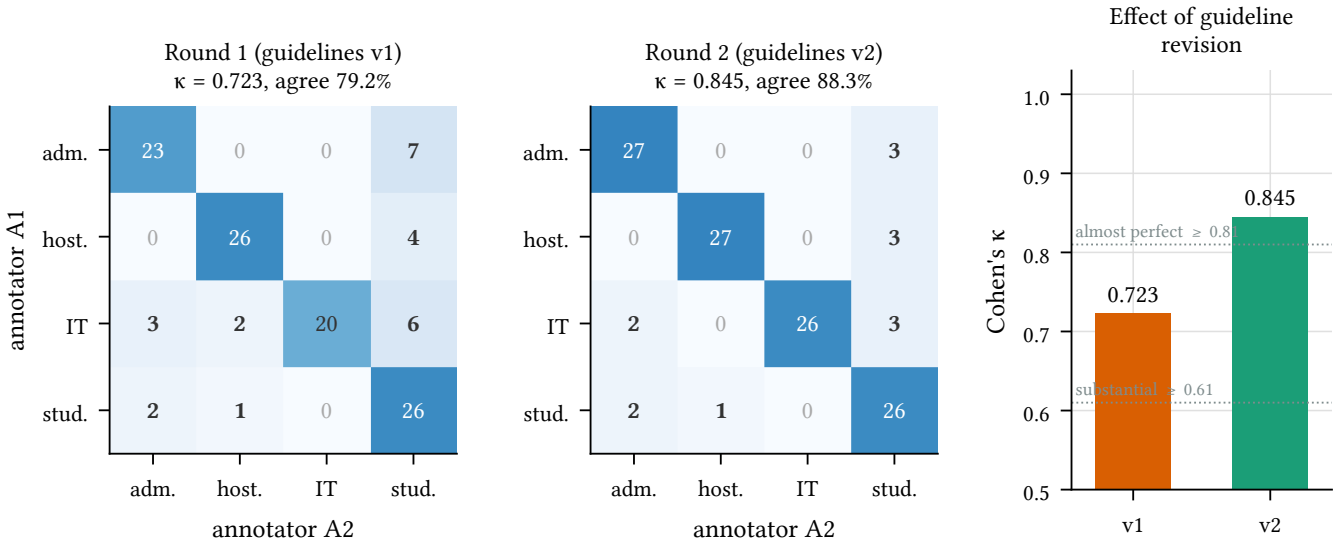


Figure 3: Intent agreement between annotators A1 and A2 on 120 double-annotated clips. Confusion matrices for guidelines v1 and v2 and the resulting Cohen's κ (recomputed from `label_studio_exports/`).

features, so its curve inherits the template-leak optimism of Section 5; the brief's decoder-confidence criterion is implemented in Assignment 2 (`retrain_pipeline.py`).

7 PART E: VERSIONING AND SPLITS

Raw audio (`data/raw/audio.dvc`), processed data, labels and metadata are versioned with DVC [3]; a local folder is the remote and the repository is tagged `a1-final`. Splits are made by *speaker* (seed 42) so no voice appears in two partitions (Table 3); Cohort B (P4, P13; Maharashtra) is held out before partitioning the remaining 17 speakers and serves as the shifted split. The leakage check passes on all pairs. File hashes are in `HANDOFF_A2.md`.

Table 3: Speaker-disjoint splits (frozen at `a1-final`).

Split	Clips	Speakers	Role
Train	171	10	model training
Val	17	3	selection, thresholds
Test = Gold	29	4	final evaluation (frozen hash)
Cohort B	32	2	shifted/drift split

8 PART F: REPRODUCIBILITY AND DATASET CARD

Rebuild. The README rebuilds the data in five commands (`git checkout a1-final`; `pip install -r code/requirements.txt`; `dvc pull`; `python code/part_e_versioning/make_splits.py`; `python code/part_b_validation/run_validation.py`). Packages are pinned in `requirements.txt` with the full freeze in `requirements_freeze.txt`; non-Python tools (Python 3.10.12, Label Studio 1.10.0, DVC 3.30.0, ffmpeg 7.1) are in `ENVIRONMENT.md`.

Models and seeds. Whisper `tiny`/`base` (official OpenAI weights) produced transcripts for augmentation benchmarking and active learning. Seeds: split 42, augmentation 123, active learning 42, Label Studio order 7.

Dataset card (summary). Sources: self-recorded helpdesk speech; size 249 clips, 75.3 min, 19 speakers; classes balanced; known biases: students only, young adult voices, few Maharashtra speakers (2), WhatsApp lossy compression upstream of the 16 kHz conversion, and the template-derived reference issue above.

9 LIMITATIONS AND CONCLUSION

The corpus is smaller than the 2 h target; annotator agreement on transcripts was inflated by template references; the augmentation WER benchmark has five clips; and the active-learning gain is not significant. The main lesson is that automated format checks cannot certify label validity: we now pair them with an audio–text alignment check and independent transcription from audio. Assignment 2 builds on the frozen splits and reports every result against both the original and the corrected gold references.

Member contributions. Table 4 lists lead responsibilities; the repository history records the commits per member.

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ACKNOWLEDGMENTS

We thank the 19 volunteer speakers and Dr. Yogesh K. Meena for the course framework. Analysis used scikit-learn, SciPy, jiwer, librosa, Matplotlib and Pandera.

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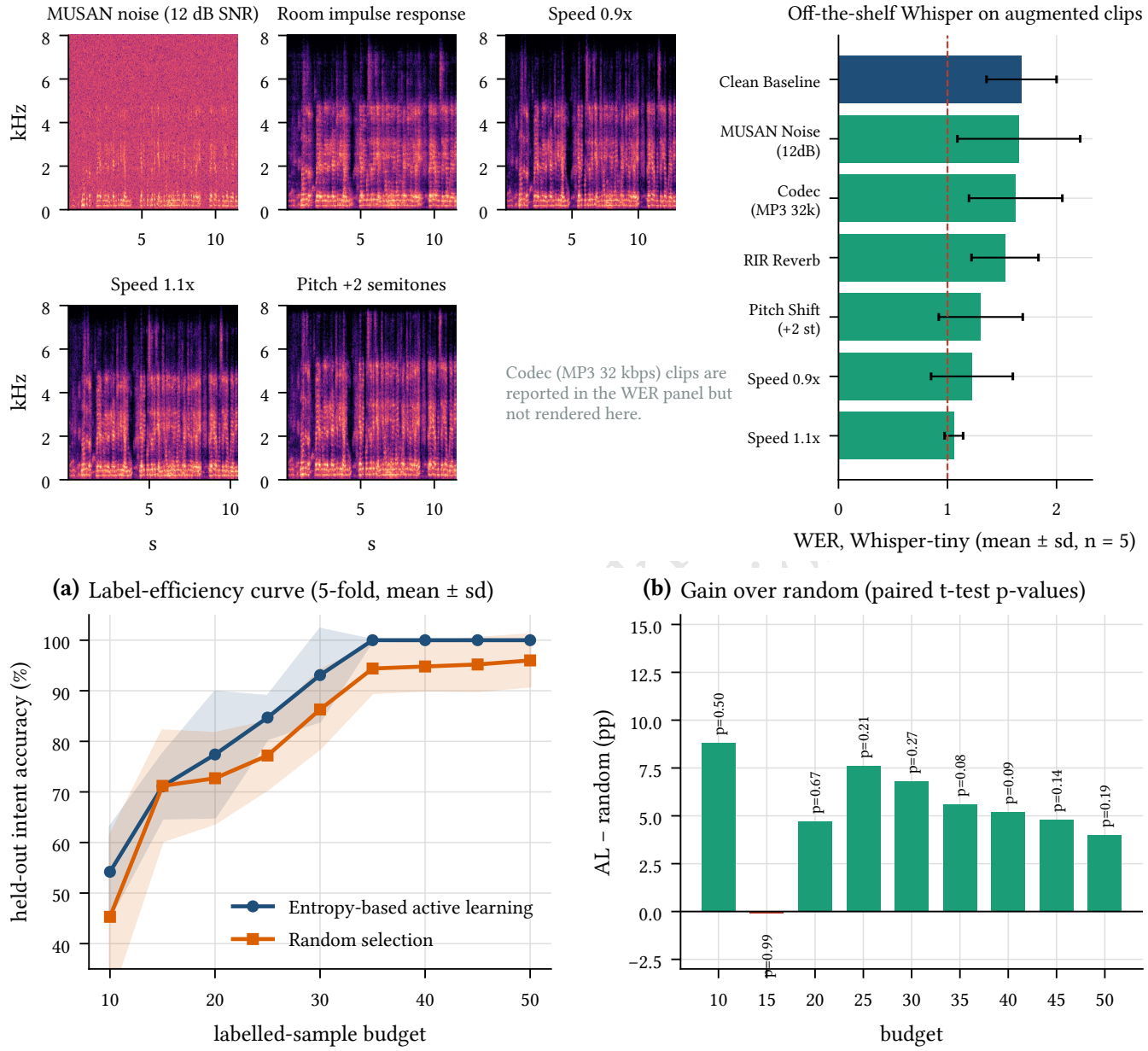


Figure 4: (a) Spectrograms of one clip under five augmentations. (b) Whisper-tiny WER on clean versus augmented audio (n=5; red line = WER of 1). (c) Label-efficiency curves, mean ± s.d. over 5 folds. (d) Active-learning gain over random with paired t-test p-values.

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A EVIDENCE INDEX

Every number in this report maps to a file in the submission zip (paths relative to the repository root).

Table 4: Contribution summary (equal 33.3% each).

Member	Lead areas
R. Vraj	**[TBD: fill in: e.g. collection pipeline, validation suite]**
V. S. Patil	**[TBD: fill in: e.g. labelling, agreement analysis, DVC]**
S. Sharma	**[TBD: fill in: e.g. augmentation, active learning, report]**

Table 5: Evidence index.

Result	Supporting file
249 clips, 75.3 min, composition	`data/metadata.csv`; `fig_a1_dataset` data
Intent balance 63/62/62/62	`data/labels/intents.csv`
11 pass / 3 fail → 14 pass	`supporting_files/validation_outputs/before.json`, `after.json`
12/12 corruptions	`validation_outputs/corruption_test.log`
κ 0.723, 0.845	`agreement_calculations/agreement_report.json`; `label_studio_exports/`
Disagreements (Table 6)	`agreement_report.json`
14/838 word overlap, gold WER 1.27	`supporting_files/predictions/production_gold.csv`
Gold corrections	`data/gold_corrections.csv`; `code/scripts/correct_gold_transcripts.py`
59/60 label-preserving	`data/processed/augmented/augmented_metadata.csv`
Whisper WER on augmented	`report/figures/augmentation_wer_table.csv`
AL vs. random, p-values	`supporting_files/plots/active_learning_results.json`
Splits and hashes	`data/splits/*.csv`; `code/HANDOFF_A2.md`
Seeds, versions	`code/ENVIRONMENT.md`; `requirements_freeze.txt`
Labelling process	`code/LABELLING_PROTOCOL.md`; guidelines v1/v2
Figure code	`code/scripts/make_report_figures.py`

Table 6: First ten of the 25 round-1 disagreements (A1 vs. A2, guidelines v1). Resolution rules are in `LABELLING_PROTOCOL.md`.

#	Clip	A1	A2
1	`P1/3_seg003`	it_support	student_services
2	`P1/3_seg005`	admissions	student_services
3	`P10/PTT-20261019-WA0020_seg002`	it_support	admissions
4	`P11/PTT-20261019-WA0015_seg001`	admissions	student_services
5	`P11/PTT-20261019-WA0015_seg003`	it_support	hostel_mess
6	`P11/PTT-20261019-WA0016_seg002`	admissions	student_services
7	`P11/PTT-20261019-WA0018_seg001`	hostel_mess	student_services
8	`P11/PTT-20261019-WA0018_seg004`	hostel_mess	student_services
9	`P12/PTT-20261019-WA0028_seg002`	student_services	admissions
10	`P13/WhatsApp_Audio_2026-10-19_`	it_support	student_services

B TEN ROUND-1 DISAGREEMENTS

C USE OF AI ASSISTANCE

AI assistants (large-language-model coding and writing assistants) were used to scaffold scripts, draft documentation and format this report. All prompts are stored verbatim in `code/prompts/` (one file per part, naming the model and the file its output went into). All numbers in this report were re-derived from result files; the kappa values and gold-set statistics were independently recomputed from raw exports and prediction files. Each member reviewed and can explain every script. **[TBD: add the exact assistant names/versions used]**

D SCREENSHOTS TO INSERT

INSERT SCREENSHOT:
`shots/labelstudio_task.png`
Label Studio task view (transcript above intent)

INSERT SCREENSHOT:
`shots/dvc_status.png`
terminal: dvc status / dvc dag

INSERT SCREENSHOT:
`shots/validation_run.png`
terminal: run_validation.py before/after

Figure 5: Evidence screenshots (place files under `report/shots/`).

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